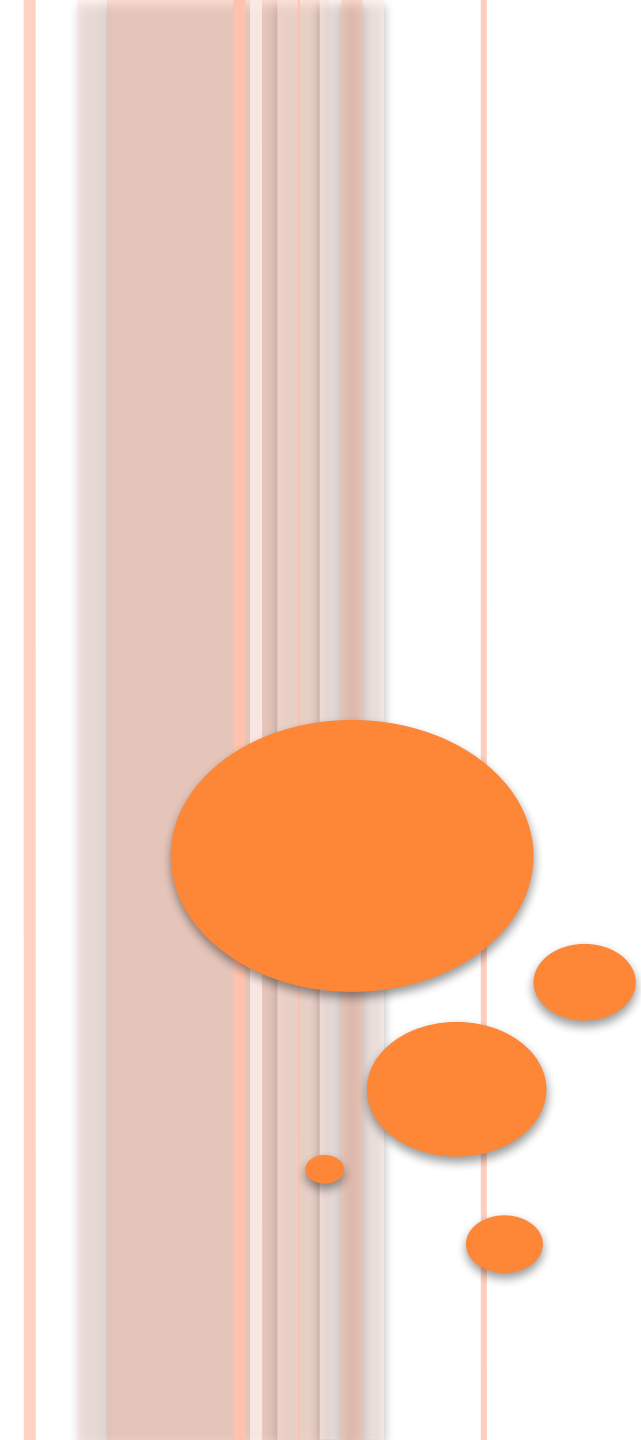




CNS PHARMACOLOGY

By Solomon A.

LEARNING OUTCOMES

- At the end of this chapters, students will be able
 - to explain how drugs work in the CNS
 - to elaborate the mechanism of CNS drugs
 - to identify the clinical use of CNS drugs
 - to explain the Sid effect and contraindications of CNS drugs

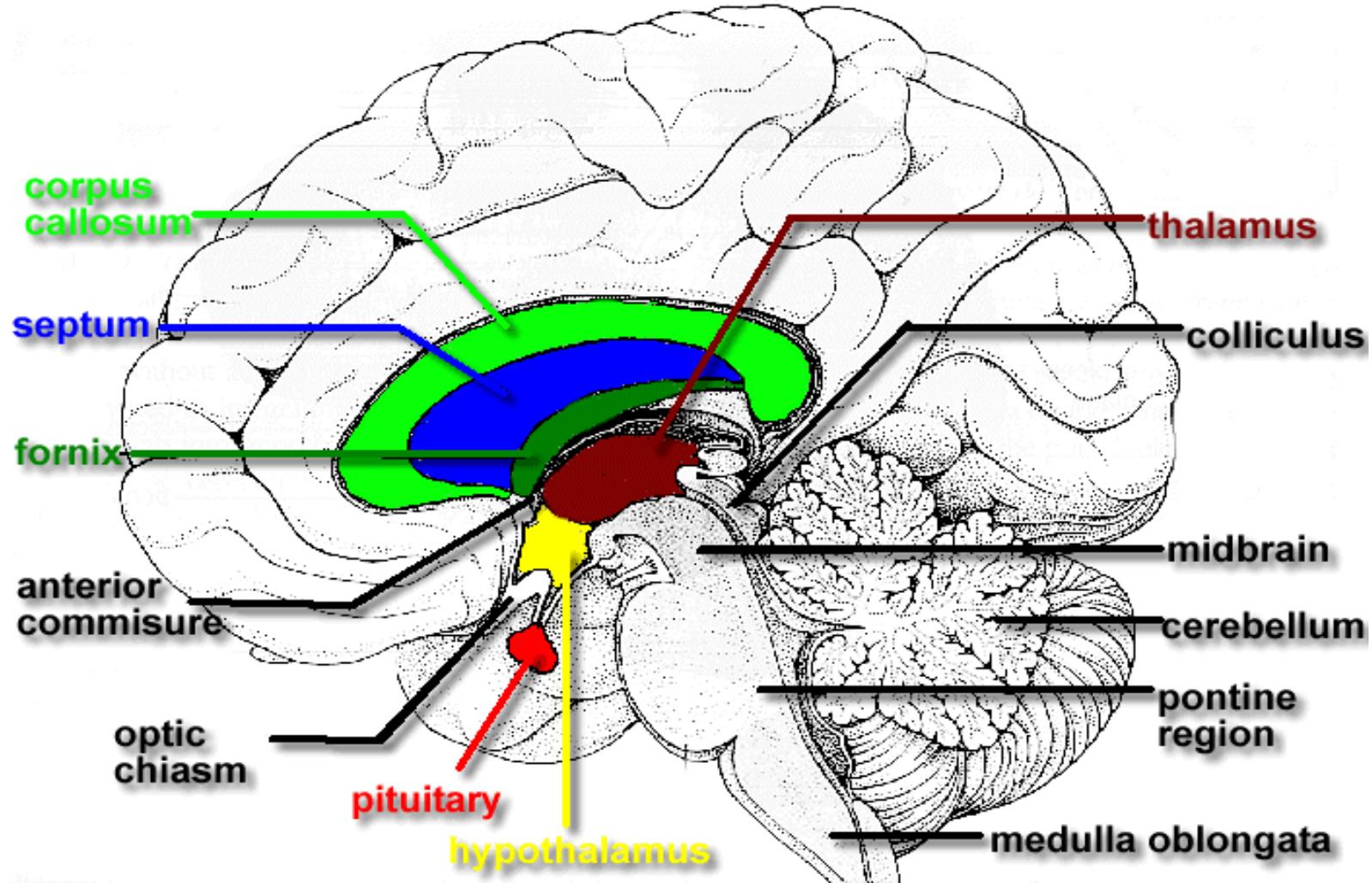


WHAT IS NEUROPHARMACOLOGY??

- **Neuropharmacology** is the study of the effects of drugs on the nervous system, aiming to develop compounds to benefit humans with psychiatric and neurological disease.



INTRODUCTION TO CNS PHARMACOLOGY



Function of the nervous system

- Receives sensory messages from the environment
- Organizes information and integrates it with already stored information
- Uses integrated information to send out messages to muscles and glands, producing organized movement and secretions
- It provides the basis for conscious experience



STRUCTURE OF CNS

1- Brain

2- Spinal cord

- The Brain formed of 3 main parts:

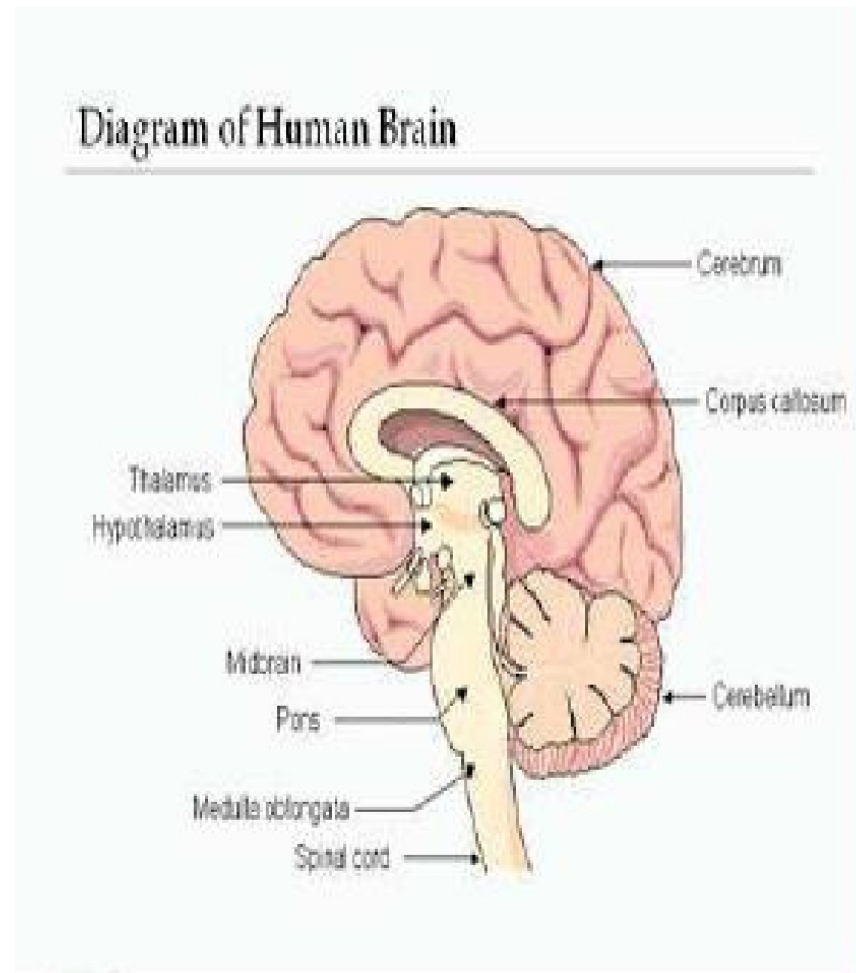
I. The Forebrain

- Cerebrum
- Thalamus
- Hypothalamus

II. The Midbrain

II. The Hindbrain

- Cerebellum
- Pons
- Medulla oblongata



❑ **Cerebrum:** Largest part of brain, Surrounded by outer cortex

❑ Cerebral cortex divided into 3 functional areas:

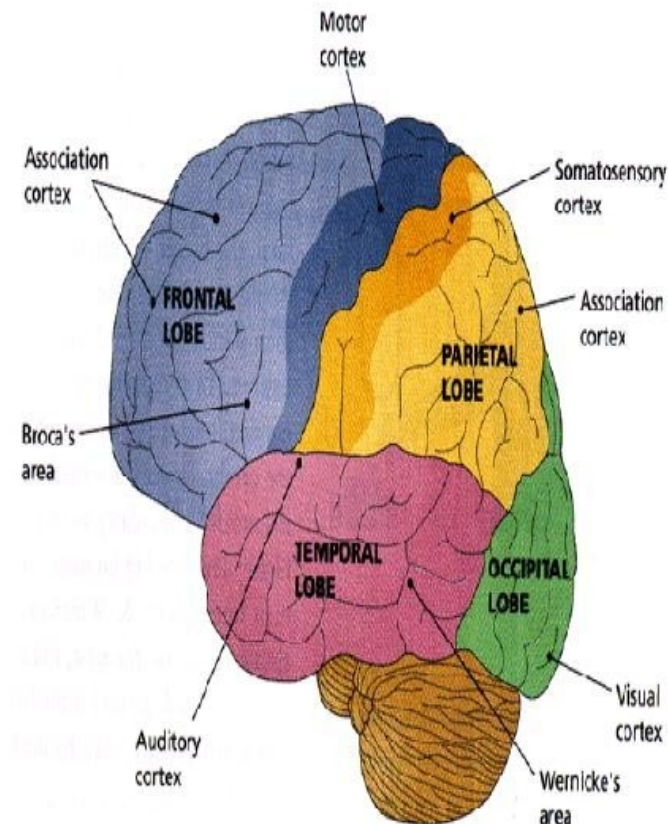
1. Sensory area (sensation)

2. Motor area

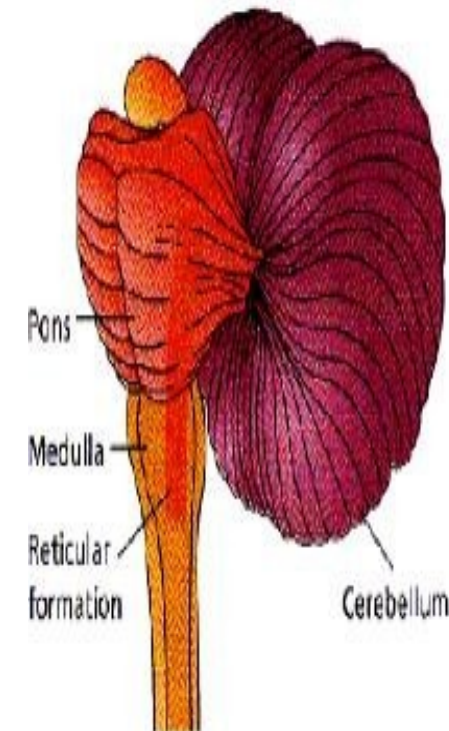
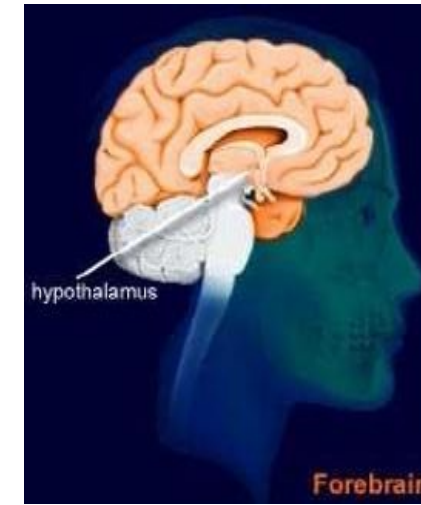
(voluntary movements) →
↑physical & mental activities, ↑↑
Convulsions

3. Association area:

- Higher mental activities as



- ❑ Thalamus: act as a relay station for **receiving N. impulses** & **sending** to sensory areas in C.C.
Contain pain center.
- ❑ The **hypothalamus**: Act as a control center for the A N S.
 - ✓ Contain heat regulating, appetite Cs.
- ❑ The mid-brain : “bridge connect cerebrum to cerebellum & Pons.
- ❑ **Cerebellum**: Resp. for balance of body & maintaining appropriate body posture & equilibrium.
- ❑ The **medulla oblongata** (M.O.) : for passage of im. bet brain & SC



❑ **Medulla oblongata** contain 5 vital Centers.[Vagal, Vasomotor, Respiratory, Cough, Vomiting(chemoreceptor trigger zone (CTZ)

II- SPINAL CORD

A cylindrical mass of nerves extends from the end of M.O. to the lower lumbar vertebrae

Impulses flow from and to brain through descending & ascending tracts of spinal cord.

Contain sex centers (Erection & ejaculation) + sweat & micturation .

Acts as a **reflex station** for receiving impulses from and to all parts of body.



CELLULAR ORGANIZATION OF THE BRAIN

- The CNS is made up of several type of specialized cells
 - Neuronal cells
 - Support cells



NEURONS

- Neurons are the highly polarized signaling cells of the brain and are classified into types based on function
 - **sensory**
 - **motor**
 - **Interneuron (Higher mental activities as consciousness, thinking, memory, behavior).**
- Neurons are electrically active cells that express a variety of **ion channels and ion transport proteins**
- The sites of interneuronal communication in the CNS are termed *synapses* functionally analogous to “junctions” ANS.



SUPPORT CELLS

- A diverse cast of support cells outnumbers neurons in the CNS.
- **Neuroglia (glia)**
 - are the most abundant support cells.
 - maintain important brain functions, such as
 - holding neurons in place
 - supplying oxygen and nutrients to neurons
 - insulating signaling between neurons
 - destroying potential pathogens.



- Neuroglia are classified as either *micro-* or *macroglia*
- The macroglia consist of
 - Astrocytes
 - Oligodendroglia
 - Ependymal cells
 - Radial glia



○Astrocytes

- are the most abundant of the glial cells
- Have **metabolic** support roles, including furnishing energy intermediates,
- Anchoring neurons to their blood supply
- regulating the external environment of the neuron by active removal of neurotransmitters and excess ions following release.



- **Oligodendroglia**

- **produce myelin**, the multilayer, compacted membranes that **electrically insulate segments of axons** and **permit nondecremental propagation of action potentials**.

- **Ependymal cells**: line the spinal cord and ventricular system and are involved in the **creation of CSF**.

- **Radial cells**: act as neuroprogenitors and scaffolds.



○ **Microglia**

- consist of specialized immune cells found within the CNS.
- act as **macrophages** to protect the neurons and are therefore mediators of immune response in the CNS.
- Microglia respond to neuronal damage and inflammation, and many diseases are associated with deficient microglia.



BLOOD-BRAIN BARRIER (BBB)

- boundary between the periphery and the CNS that forms a permeability barrier to the passive diffusion of substances from the bloodstream into the CNS.
- An exception exists for lipophilic molecules, which diffuse fairly freely across the BBB and accumulate in the brain.



CNS NEUROTRANSMITTERS

- Neurotransmitters can be classified by chemical structure into various categories, including
 - *Amino acids*
 - *Acetylcholine*
 - *Monoamines*
 - *Neuropeptides*
 - *Purines*
 - *Gases*



CNS Neurotransmitters...

- Neurotransmitters in the CNS are either inhibitory or Excitatory
- **Inhibitory**
 - GABA
 - Glycine
 - Monoamines including Dopamine
- **Excitatory**
 - Glutamate
 - Aspartate
 - Acetylcholine



- ❑ A nerve impulse (electric current) passed along axon to presynaptic membrane.
- ❑ Release neurotransmitter into synaptic cleft.
- ❑ NT interacts with receptors on effector cells to induce response.
- ❑ NT released in response to action potentials is voltage dependent & require calcium influx (Neuroregulators).

❑ **Transmission in C.N.S. occurs in 2 ways:**

A- Release of Excitatory transmitter by Neuron

- (1) Cause depolarization of postsynaptic membrane of neuron
- (2) Cause Conduction of N impulse [**postsynaptic excitation**].

B- Release of Inhibitory Transmitter by Neuron

- (1) Cause Hyper polarization of postsynaptic membrane .
- (2) Block conduction of nerve impulse [**postsynaptic inhibition**].



TERMINATION OF NEUROTRANSMITTER ACTION

- There are two primary mechanisms for terminating the signaling of released transmitters.
- **conversion** of the transmitter into an inactive compound via an enzymatic reaction. E.g. AchE
- **clearance** of the neurotransmitter by transport proteins present on presynaptic neurons E.g. NET, SERT and DAT
- ❖ slow diffusion of the transmitter away from the synapse and subsequent degradation



NEUROTRANSMISSION SYSTEMS CONT...

- Major neurotransmission systems are the;
 - Cholinergic,
 - Dopaminergic,
 - GABA-ergic,
 - Noradrenergic, and
 - Serotonergic networks.

The cholinergic system:

- Uses acetylcholine (Ach) as its NT.
- Ach,
 - the first substance to be designated as a NT in the CNS.
- Located in many areas of the brain,
 - especially high concentrations in the **motor cortex** and **basal ganglia**.
- **Exerts excitatory effects at synapses** and **NMJs** and **inhibitory effects** at some sites.
- In the CNS, acetylcholine is associated with,
 - **arousal, learning, memory, motor conditioning, and speech.**
- **Dementia and Parkinsonism** is associated with abnormalities in cholinergic pathways

The dopaminergic system:

- Uses dopamine as its NT.
- Originate primarily **from substantia nigra, ventral tegmental area (VTA) and hypothalamus.**
- Projects to different areas of brain including **striatum, limbic areas (e.g. amygdala, hippocampus, nucleus accumbens)**, frontal and prefrontal lobe cortex, pituitary gland.
- Dopamine is important in;
 - motor control (Parkinsonism is due to dopamine deficiency),
 - behavioural effects (excessive dopamine activity is implicated in schizophrenia),
 - hormone release (inhibits prolactin secretion) and
 - chemoreceptor trigger zone causes nausea and



- Two groups of dopamine receptors have been identified.
- One group includes D1 and D5 receptors,
 - **activate** adenylyl cyclase to produce cAMP.
- The other group includes D2, D3, and D4 receptors.
 - **inhibit** activation of adenylyl cyclase,
 - suppress calcium ion currents, and
 - activate potassium ion currents.

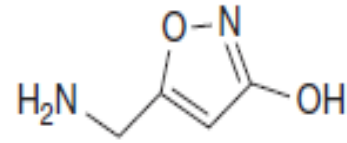
NEUROTRANSMISSION SYSTEMS

CONT...

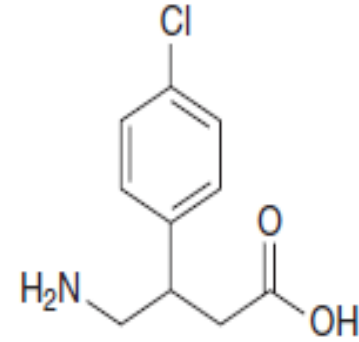
- **The GABA-ergic system**
 - uses GABA as its neurotransmitter.
- GABA
 - **Major inhibitory** NT of CNS
 - Found in virtually every region of the brain.
- GABA receptors- two main types, A and B.
- GABA_A receptor:
 - a chloride ion channel that opens when GABA is released from presynaptic neurons.
 - Activation causes **hyperpolarization**.
- GABA_B receptor:
 - Also leads to **decreased presynaptic Ca²⁺ influx**..
 - Leads to increased efflux of K⁺ and hyperpolarization



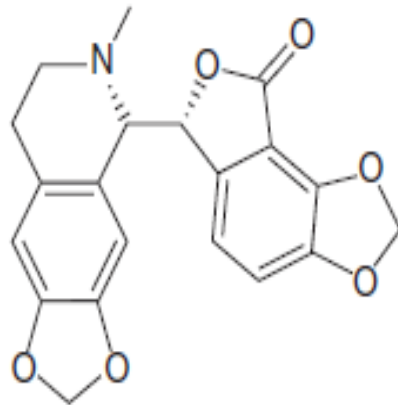
GABA
(γ -aminobutyric acid)



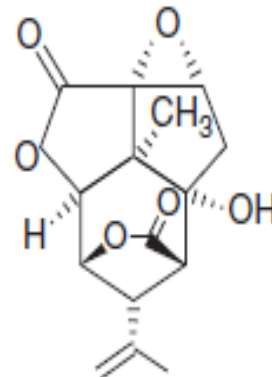
MUSCIMOL
(GABA_A agonist)



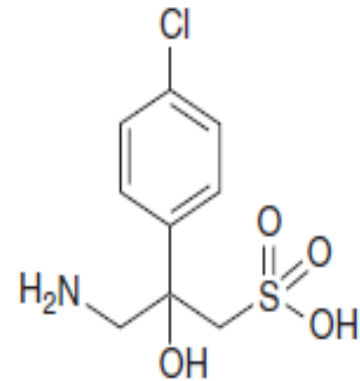
BACLOFEN
(GABA_B agonist)



BICUCULLINE
(GABA_A antagonists)

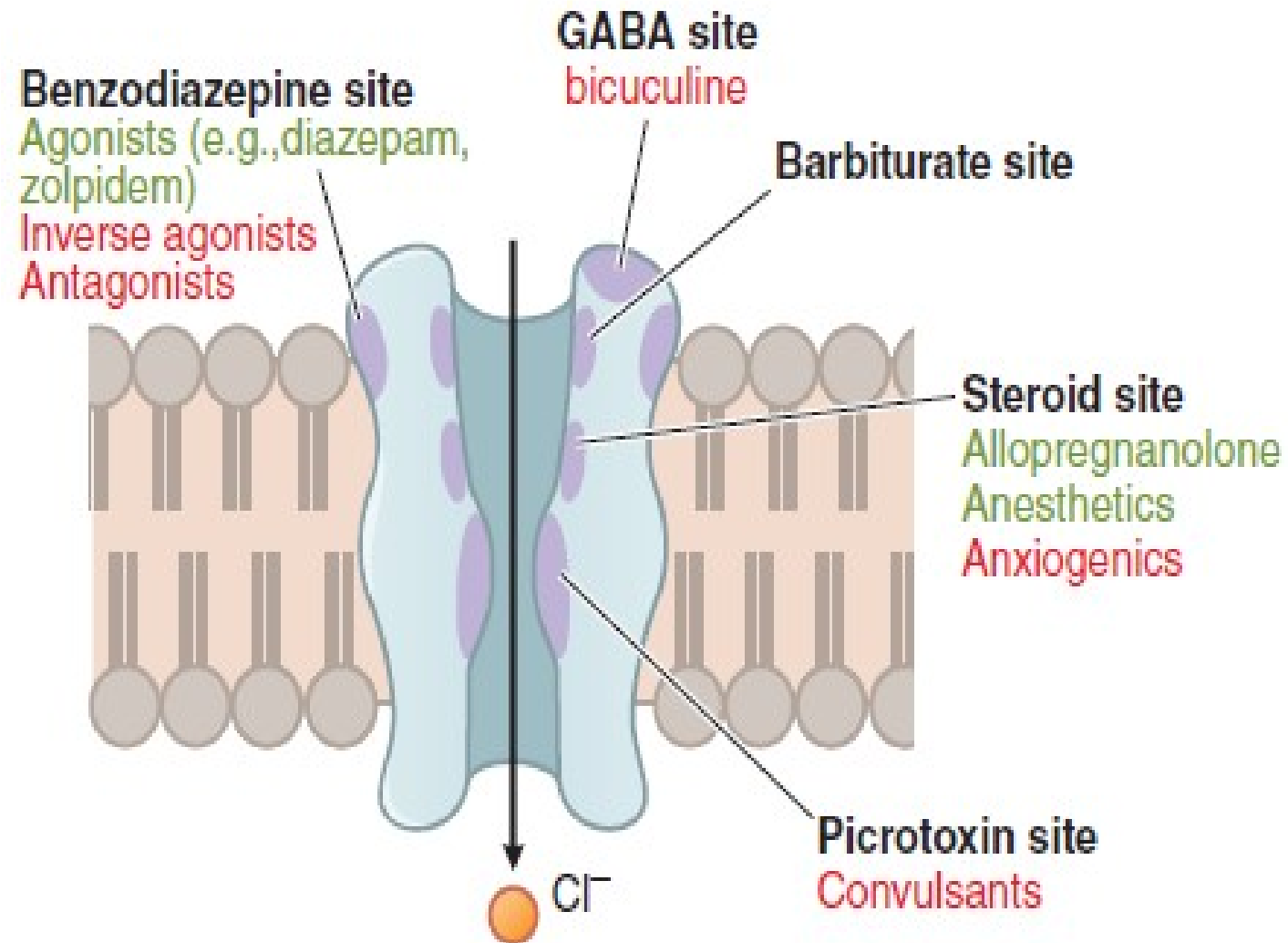


PICROTOXIN



2-OH-SACLOFEN
(GABA_B antagonist)





NEUROTRANSMISSION SYSTEMS CONT...

The Noradrenergic system:

- uses norepinephrine as its NT
- Originate primarily in **locus coeruleus** (in the pons).
- Projects diffusely to cortex and brainstem.
- extends to virtually every area of the brain.
- Noradrenergic system is associated with;
 - **mood, wakefulness and alertness, reward, Arousal, Attention,**
- Control mood
 - Functional deficiency contribute to **depression.**
 - Function of reward system.
 - Cocaine /

THE NORADRENERGIC SYSTEM CONT...

- Norepinephrine receptors in the CNS, as in the sympathetic nervous system,
 - divided into alpha- and beta-adrenergic receptors and their subtypes.
- Activation of $\alpha 1$, $\beta 1$, and $\beta 2$ receptors:
 - **Adenyl cyclase and the production of cAMP.**
- Activation of $\alpha 2$ receptors:
 - associated with decreased **production of cAMP**
- However, the effects of $\alpha 2$ receptor are thought to stem mainly from;
 - activation of receptor-operated potassium ion channels and
 - suppression of voltage-operated calcium ion channels

THE NORADRENERGIC SYSTEM CONT...

α 2 receptors on the presynaptic nerve ending
– regulate norepinephrine release.



NEUROTRANSMISSION SYSTEMS CONT...

The serotonergic system:

- uses serotonin (5-hydroxytryptamine or 5-HT) as its NT.
- Serotonin-synthesizing neurons
 - widely distributed in the CNS, beginning in the midbrain (**raphe nuclei**) and **projecting into the thalamus, hypothalamus, cerebral cortex, and spinal cord.**
 - synthesized from the amino acid tryptophan.
- Serotonin in CNS is associated with,
 - **Mood, Arousal** (sleep–wake cycle), **emotional behavior, temperature regulation, inhibition of pain pathways** in the spinal cord.
 - **Migraine**

THE SEROTONERGIC SYSTEM

CONT...

Serotonin receptors

- Many types and subtypes
 - 5-HT₁₋₇
- Mostly metabotropic
 - Except 5-HT₃, ionotropic
- 5-HT_{1B/D}
 - Presynaptic **autoreceptors**
 - Inhibitory

NEUROTRANSMISSION SYSTEMS

CONT...

The amino acid system

- Includes several amino acids that may serve as both structural components for protein synthesis and NTs.
- Amino acids were recognized as NTs relatively recently,
 - their roles and functions in this regard have not been completely elucidated.
- Aspartate:
 - an excitatory NT found in high concentrations in the brain.
- Aspartate and glutamate are considered the major fast-acting, excitatory NTs in the brain.

GLYCINE

- Glycine act as an **inhibitory neurotransmitter**, particularly in the spinal cord and brainstem.

- terminated by reuptake through specific transporters (**GLYT1 and GLYT2**).

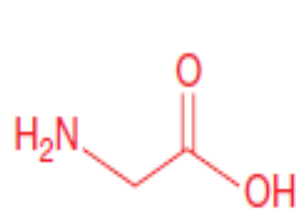
- linked to Cl^- channels and are prominent in the **brainstem and spinal cord**.

- **Taurine and β -alanine** are agonists of glycine receptors;

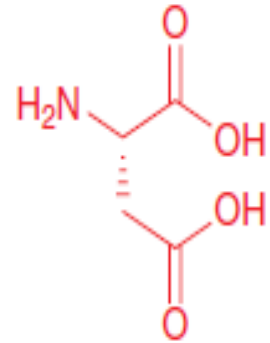
strychnine, a potent convulsant, is a selective antagonist



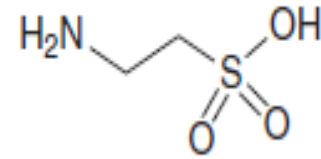
GLYCINE...



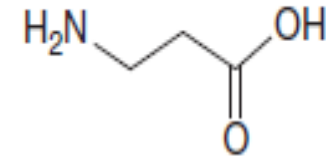
GLYCINE



ASPARTIC ACID

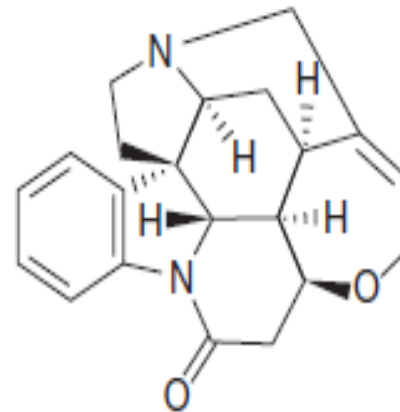


TAURINE



β -ALANINE

(Glycine receptor agonists)



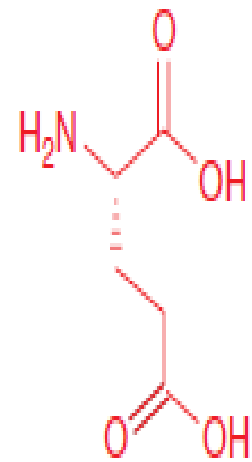
STRYCHNINE
(Glycine antagonist)



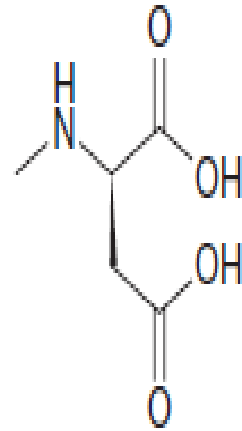
THE AMINO ACID SYSTEM CONT...

- **Glutamate**
 - **considered the most important excitatory NT in the CNS.**
 - **It occurs in high concentrations in virtually every area of the CNS, including the cerebral cortex, basal ganglia, limbic structures, and hippocampus.**
- **Several subtypes of glutamate receptors have been identified.**
- **The functions of these receptor subtypes have not**

- Almost all neurons that have been tested are strongly excited by glutamate.
- This excitation is caused by the activation of both ionotropic and metabotropic receptors.
- The ionotropic receptors can be further divided into three subtypes based on the action of selective agonists:
 - ✓ α -amino-3-hydroxy-5-methylisoxazole-4- propionic acid(AMPA),
 - ✓ kainic acid(KA), and
 - ✓ N-methyl-D-aspartate (**NMDA**).

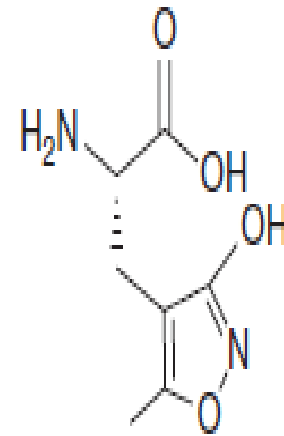


GLUTAMIC ACID



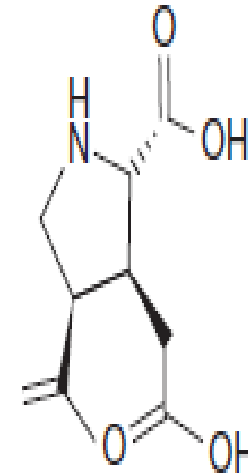
NMDA

(*N*-methyl-*D*-aspartate)



AMPA

(*α*-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid)

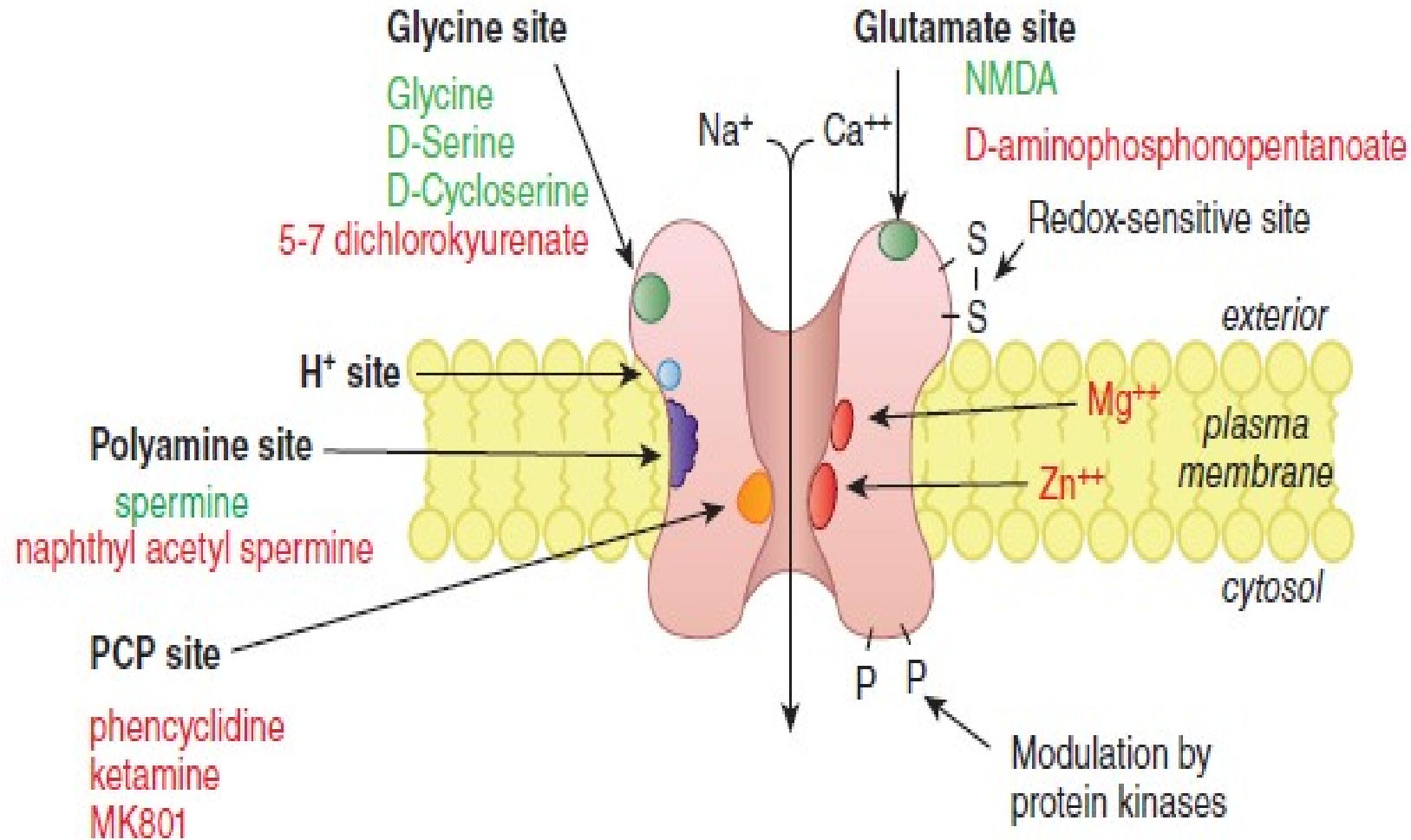


KAINIC ACID

(Glutamate receptor subtype-specific agonists)



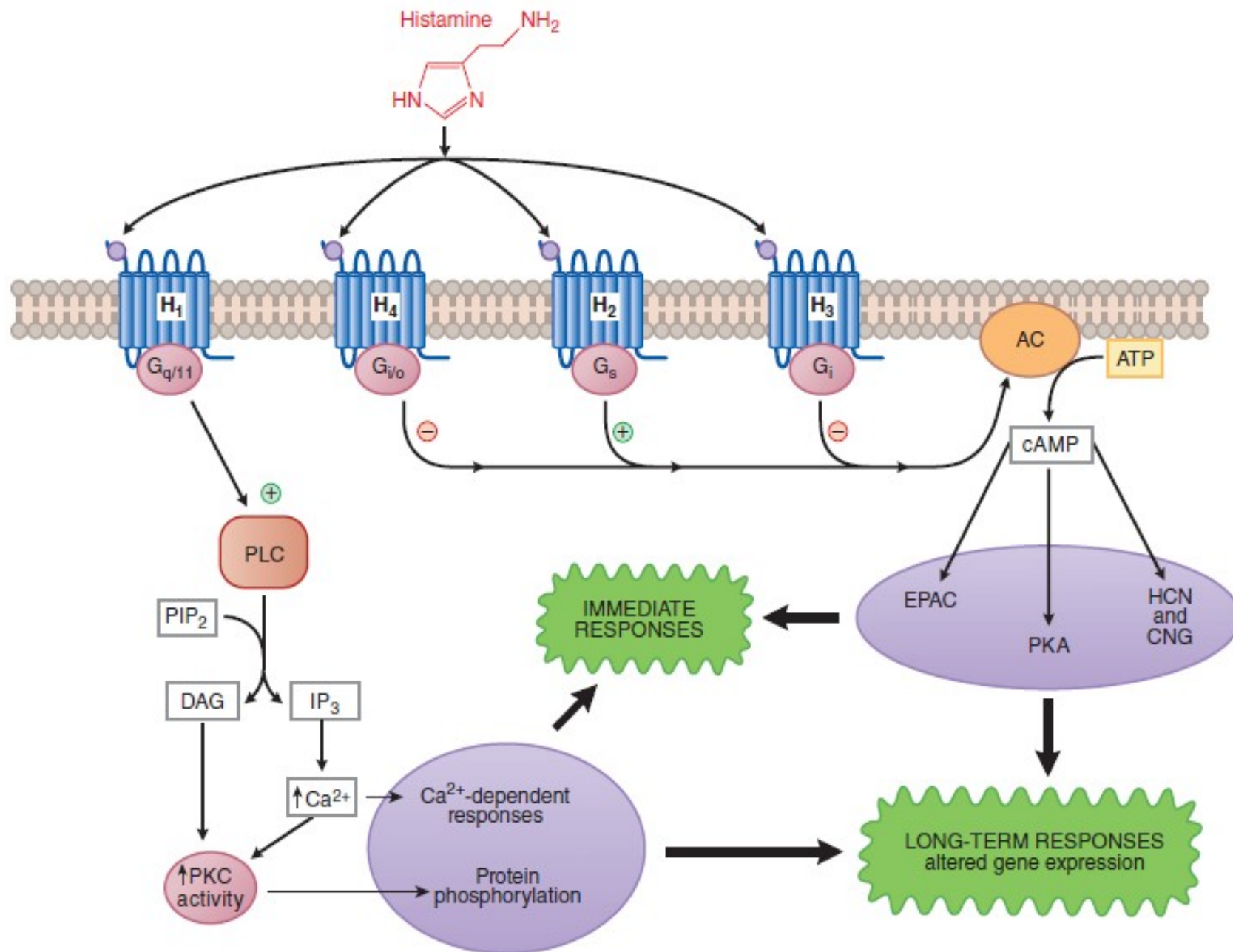
PHARMACOLOGIC BINDING SITES ON THE NMDA RECEPTOR



Binding of both glutamate and glycine is necessary for activation

HISTAMINE

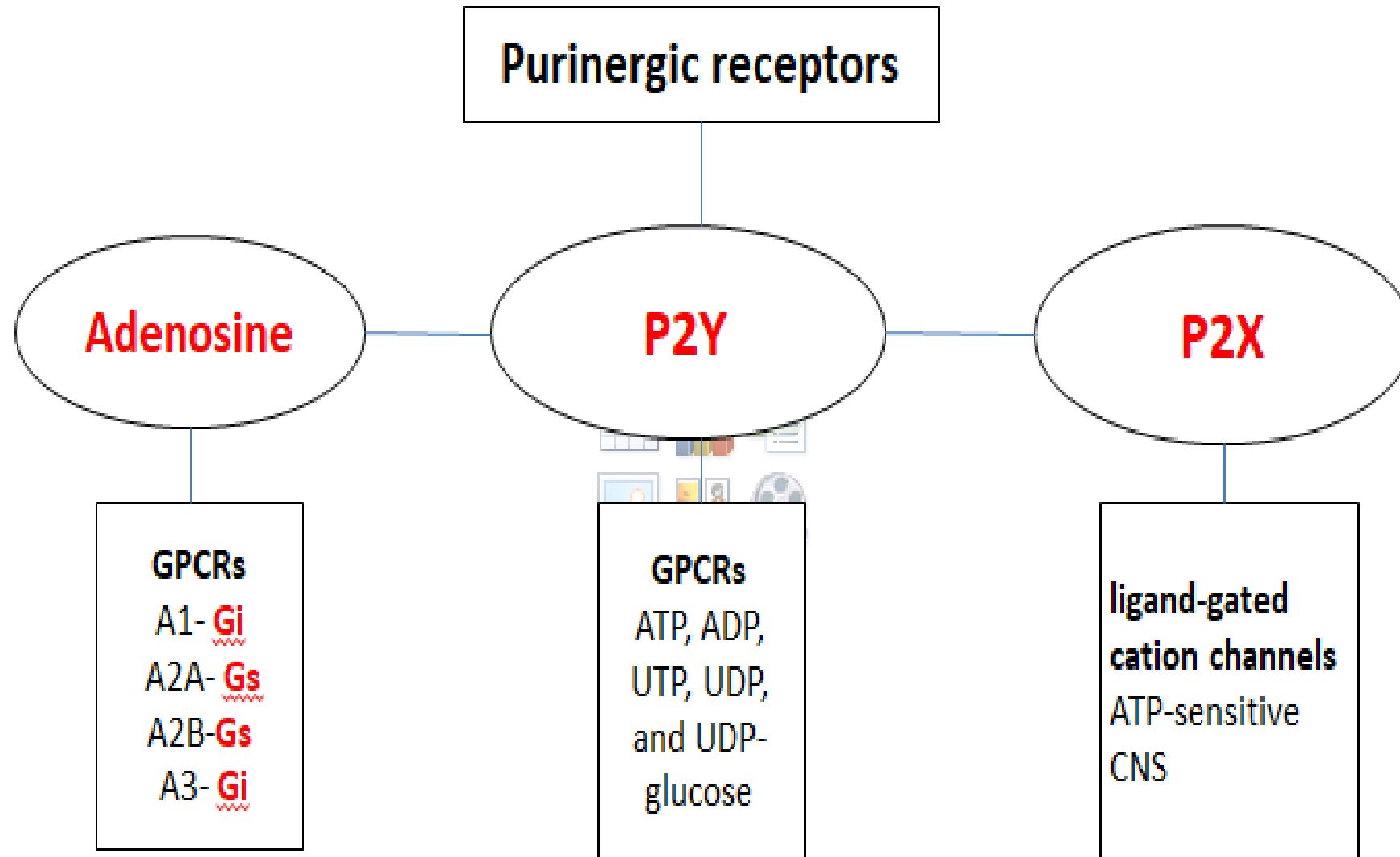
- Histaminergic neurons are located in the ventral posterior hypothalamus
- It is thought to affect arousal, body temperature, and vascular dynamics.
- Histamine signals through four GPCR subtypes (H1–H4) that regulate either adenylyl cyclase or PLC
 - H1 receptors are widely **distributed in the brain**
 - H2 receptors activate adenylyl cyclase and are primarily involved in **gastric acid secretion and smooth muscle relaxation**
 - H3 receptors are also present in the CNS and can act as **autoreceptors** on histaminergic neurons
 - H4 receptors are expressed on cells of **hematopoietic origin**



PURINES

- Adenosine, ATP, UDP, and UTP have roles as extracellular signaling molecules.
- ATP is also a component of many neurotransmitter storage vesicles and is released along with transmitters.
- Extracellular nucleotides and adenosine can act on a family of diverse purinergic receptors,
 - involved in memory and learning, locomotor behavior, and feeding.

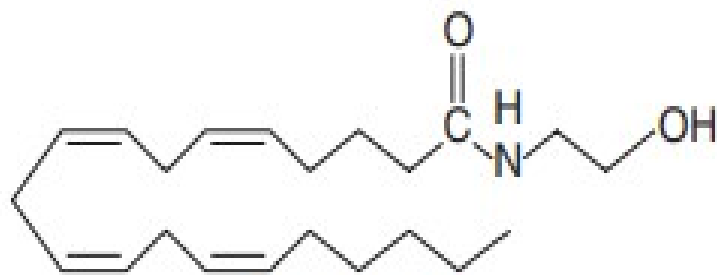




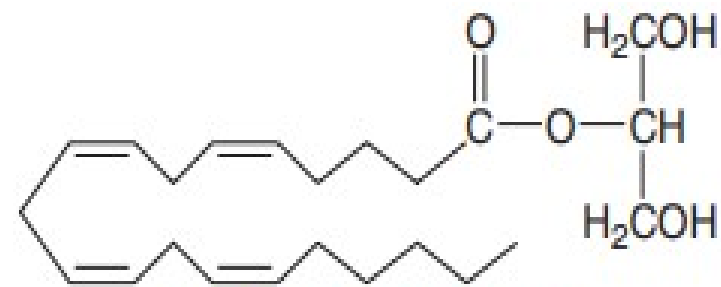
CANNABINOIDS

- THC was identified as a **psychoactive substance in marijuana**.
- Two cannabinoid receptors and the identification of endogenous compounds that modulate them.
- The two receptor subtypes
 - CB1-brain
 - CB2-immune cells
- are GPCRs that couple to Gi/Go to inhibit adenylyl cyclase

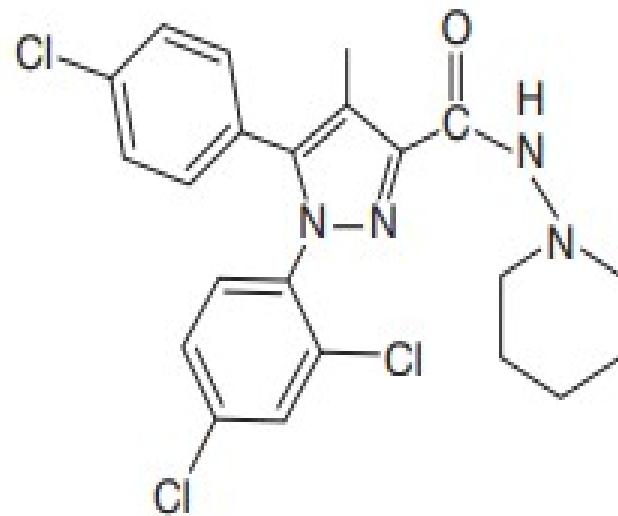




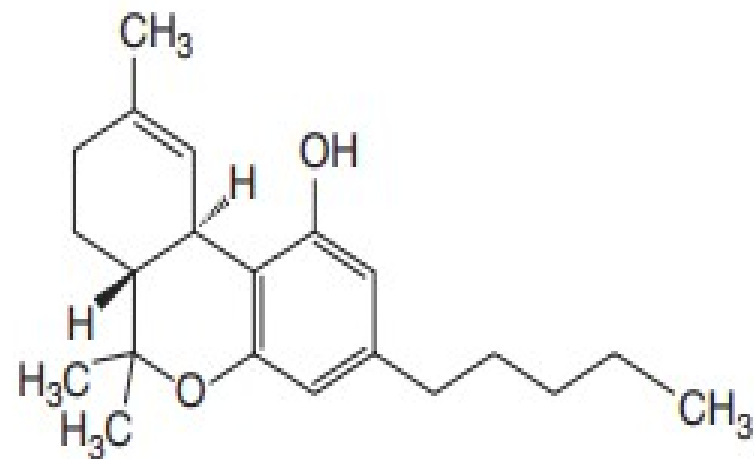
Anandamide



2-Arachidonylglycerol



Rimonabant



Δ⁹-Tetrahydrocannabinol



NT CHANGES...

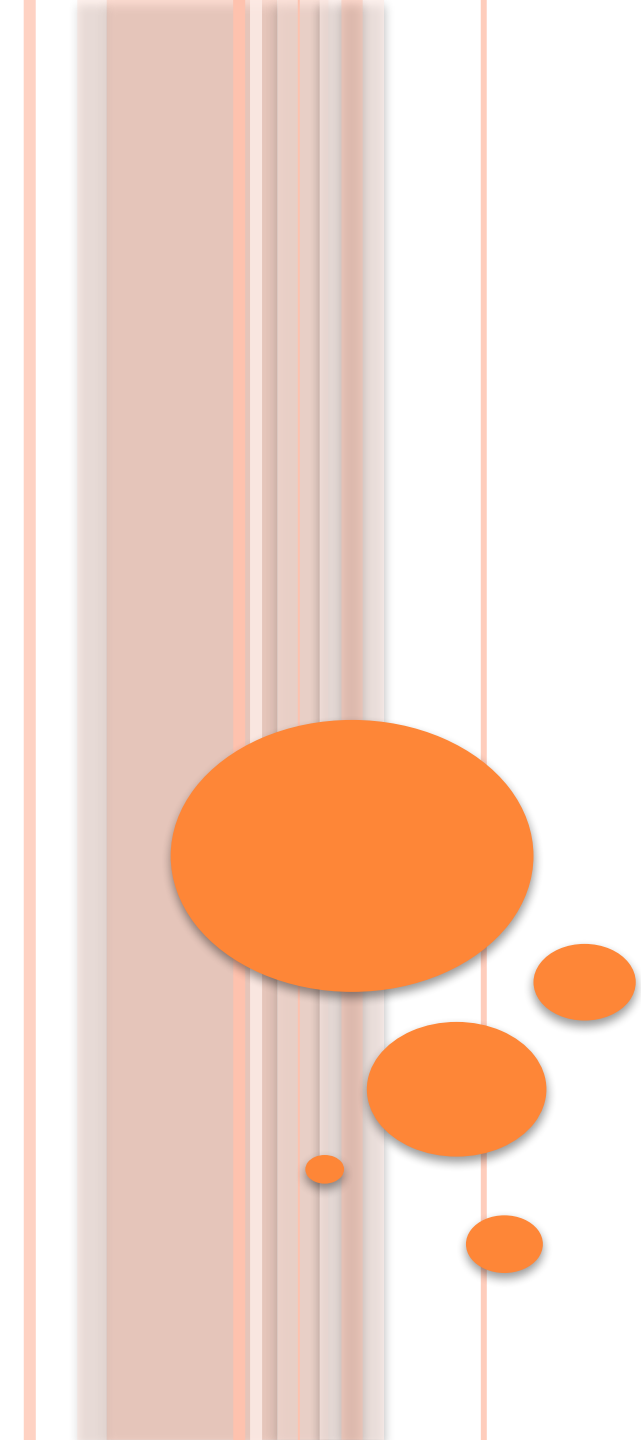
- Parkinson's disease:
↓ Dopamine (relatively ↑ Acetylcholine)
- Depression:
↓ Serotonin, ↓ NA
- Schizophrenia:
↑ Dopamine



INTRODUCTION TO CNS DRUGS

- Nearly all drugs with CNS effects act on specific receptors that modulate synaptic transmission.
- A very few agents such as general anesthetics and alcohol may have nonspecific actions on membranes, but even these non–receptor-mediated actions result in demonstrable alterations in synaptic transmission.



A decorative vertical bar on the left side of the slide, featuring a light orange background with several thin, vertical orange lines. To the right of this bar, there are five orange circles of varying sizes arranged in a cluster.

HYPNOTICS AND SEDATIVES

INTRODUCTION

- **A sedative drug** decreases activity, moderates excitement, and calms the recipient.
- **Hypnotic drug** produces drowsiness and facilitates the onset and maintenance of a state of sleep that resembles natural sleep in its electroencephalographic characteristics.



IDEAL HYPNOTIC

- Should cause a transient decrease in the level of consciousness for the purpose of sleep without lingering effect.
- Have no potential for decreasing or arresting respiration
- Produce no abuse, addiction, tolerance or dependence



USES

1. Antianxiety agents in emotional strain
2. Anticonvulsant
3. Muscle relaxants
4. General Anaesthetics
5. Potentiation of analgesic agents
6. Adjuvants to anaesthesia
7. In hypertension



○ **Barbiturates:**

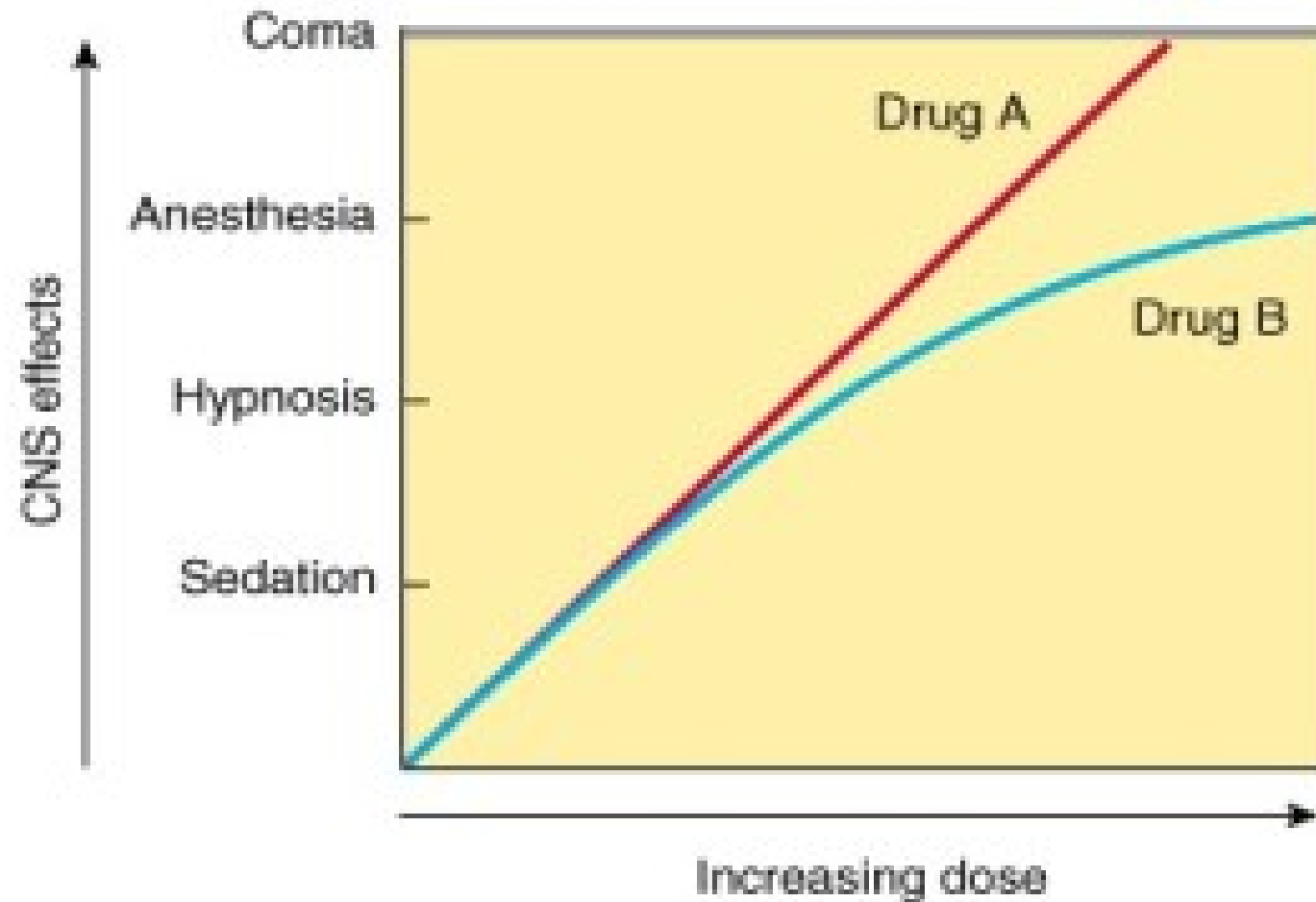
- **Long acting:** Phenobarbitone
- **Short acting (Intermediate):** Pentobarbitone, Butobarbitone, Amylobarbitone
- **Ultra-short acting:** Methohexitone, Thiopentone, Secobarbitone

○ **Benzodiazepines**

- **Newer Non-benzodiazepines:** Zopiclone, Zolpidem, Zaleplon etc. (Chloral hydrate, paraldehyde, meprobamate – not used anymore)







- The linear slope for drug A is typical of many of the older sedative-hypnotics, including the barbiturates and alcohols
- Deviations from a linear dose-response relationship for drug B – proportionately greater dosage increments to achieve central nervous system depression



- **Agonists**

- **GABA**: promotes Cl influx
- **Barbiturates**: facilitates & mimics GABA action
- **Benzodiazepines**: facilitates GABA action
- Alcohol, Inhalational anaesthetics, propofol: open Cl channel directly

- **Antagonists**

- **Bicuculine**: competitive antagonist at GABA Rc
- **Flumazenil**: competitive antagonist at BZD site
- **Picrotoxin**: blocks Cl channel non-competitively

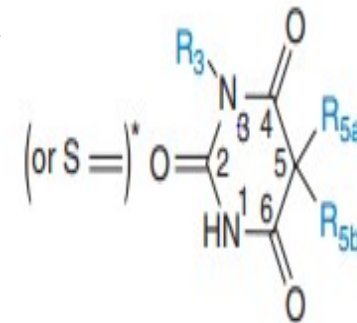
- **Inverse agonists**

- **B-carbolines** (DMCM-dimethoxyethyl-carbomethoxy-B-carboline):inverse



BARBITURATES

- Last century – up to 1960 – not used majorly now – but prototype drug
- Chemically – condensation of malonic acid and urea – Barbituric acid
- Important drawbacks:
 - General depressants of all excitable cells (C
 - Potentially Fatal Respiratory Depression
 - narrow therapeutic range
 - Potent liver inducers: interactions



*O except in thiopental, where it is replaced by S.

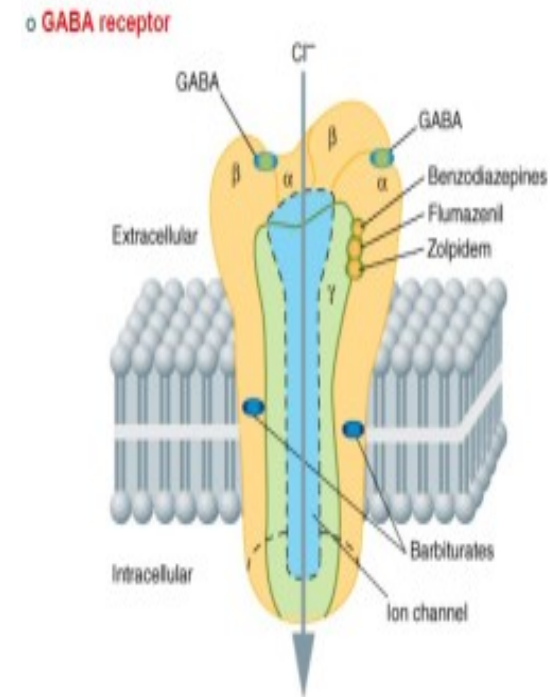
- Dose dependent effect: sedation – sleep – anaesthesia – coma
- Phenobarbitone – higher anticonvulsant :
sedative ratio

Mechanism of action

- Act primarily at **GABA receptor-Cl⁻ channel**
- **Potentiates GABAergic inhibition by increasing the lifetime of Cl⁻ channel opening**



- Bind to other site than GABA –
located in α and β subunit
- Also enhance BZD binding
- **High doses** – directly increases Cl^- conductance (GABA mimetic) and inhibit Ca^{++} dependent release of neurotransmitters
- Also depress **Glutamate induced neuronal depolarization**
- Very High doses – **depress voltage sensitive Na^+ and K^+ channel**



- **Respiration:** Depressed – neurogenic, hypercapneic and hypoxic drives to **respiratory center depressed**
- **CVS:** Slight decrease in BP and Heart Rate
 - **Toxic doses:** Marked fall in BP and HR – due to ganglion blockade, vasomotor center depression and direct cardiac action – reflex tachycardia
- **Skeletal Muscle:** Little effect but **anaesthetic doses reduce muscle contraction**
- **Smooth muscle:** Tone and motility of bowel is decreased.
- **Kidney:** Reduce urine flow – due to decrease BP and ADH release



PK

- Well absorbed orally – widely distributed
- **Entry** into CNS depends on **lipid solubility**
- **Plasma protein binding** – Phenobarbitone – 20%, Thiopentone – 75%
- **Crosses placenta** and secreted in milk
- **Metabolism:** metabolized in liver by **oxidation, dealkylation and conjugation**
- **Excretion:** Long acting – excreted significantly **unchanged** - **Alkalinization – excretion**
- **Microsomal enzyme inducer** – induce own metabolism and Others



- For relief of **anxiety**
- For insomnia
- For **sedation and amnesia** before and during medical and surgical procedures
- For treatment of **epilepsy and seizure states**
- As a component of **balanced anesthesia** (intravenous administration)
- For control of **ethanol or other sedative-hypnotic withdrawal states**
- For **muscle relaxation in specific neuromuscular disorders**

- **Adverse Effects:** Hang over, tolerance and dependence, mental confusion, impaired performance – accidents
 - **Idiosyncrasy:** Occasional excitement – In elderly
 - **Hypersensitivity:** Rashes, swelling of eyelids, lips etc.
 - **Tolerance and dependence:** Cellular and pharmacokinetic – on repeated use
 - Physical and psychological dependence – abuse liability
- **Drug Interactions:**
 - **Induce** metabolism of many drugs – warfarin, steroids, OCP, chloramphenicol, tolbutamide
 - Alcohol, antihistamines, opioids – **CNS depression**
 - Sodium valproate – **increases plasma conc. Of phenobarbitone**
 - Phenobarbitone – **competitively induces phenytoin metabolism**



ACUTE BARBITURATE POISONING

- Mostly suicidal – **excessive CNS depression – flabby and comatose** with shallow and failing **respiration**, **CVS collapse**, **renal shut down**, **pulmonary complications**
- **Treatment:**
 - Gastric lavage (activated charcoal)
 - Supportive – patent airway, assisted respiration, oxygen, IV fluid and **vasopressors** like **Dopamine**
 - **Alkaline diuresis:** **Sodium bicarbonate 1 meq/kg IV** with or without **mannitol** – for long acting ones
 - **Haemodialysis:** **highly effective** in long as well as short acting ones



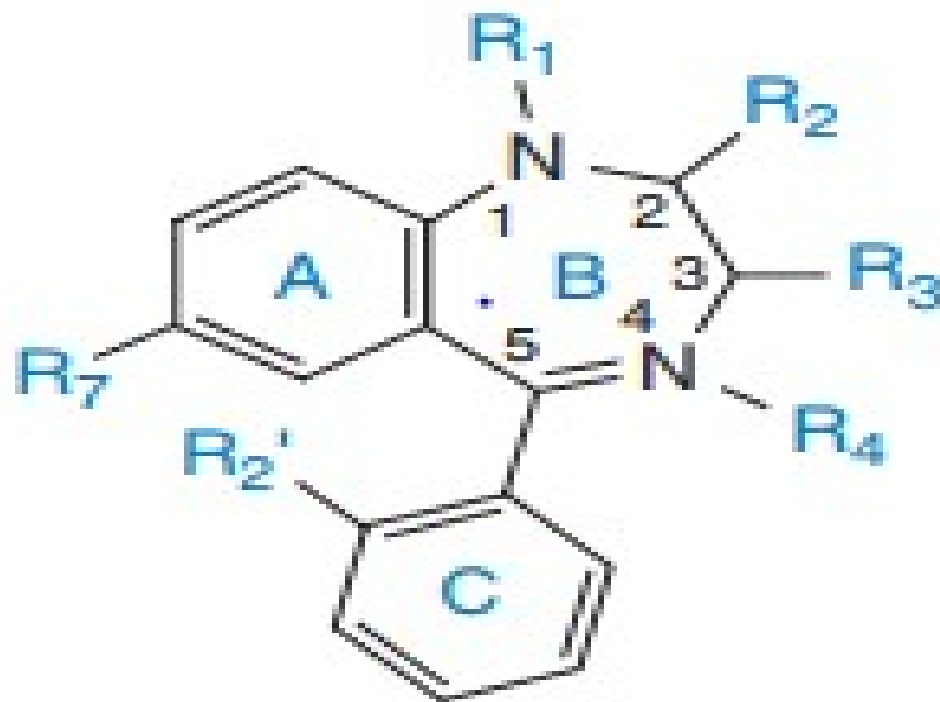
BENZODIAZEPINES

- All benzodiazepines in clinical use **promote the binding** of the major inhibitory neurotransmitter GABA to the GABA-A receptor, a pentameric ligand-gated, anion-conducting channel.
- Considerable heterogeneity exists among human GABA-A receptors; this heterogeneity is thought to contribute to the myriad effects of these agents *in vivo*.



Hypnotic	Antianxiety	Anticonvulsant
Diazepam	Diazepam	Diazepam
Flurazepam	Chlordiazepoxide	Lorazepam
Nitrazepam	Oxazepam	Clonazepam
Alprazolam	Lorazepam	Clobazam
Temazepam and Triazolam	Alprazolam	





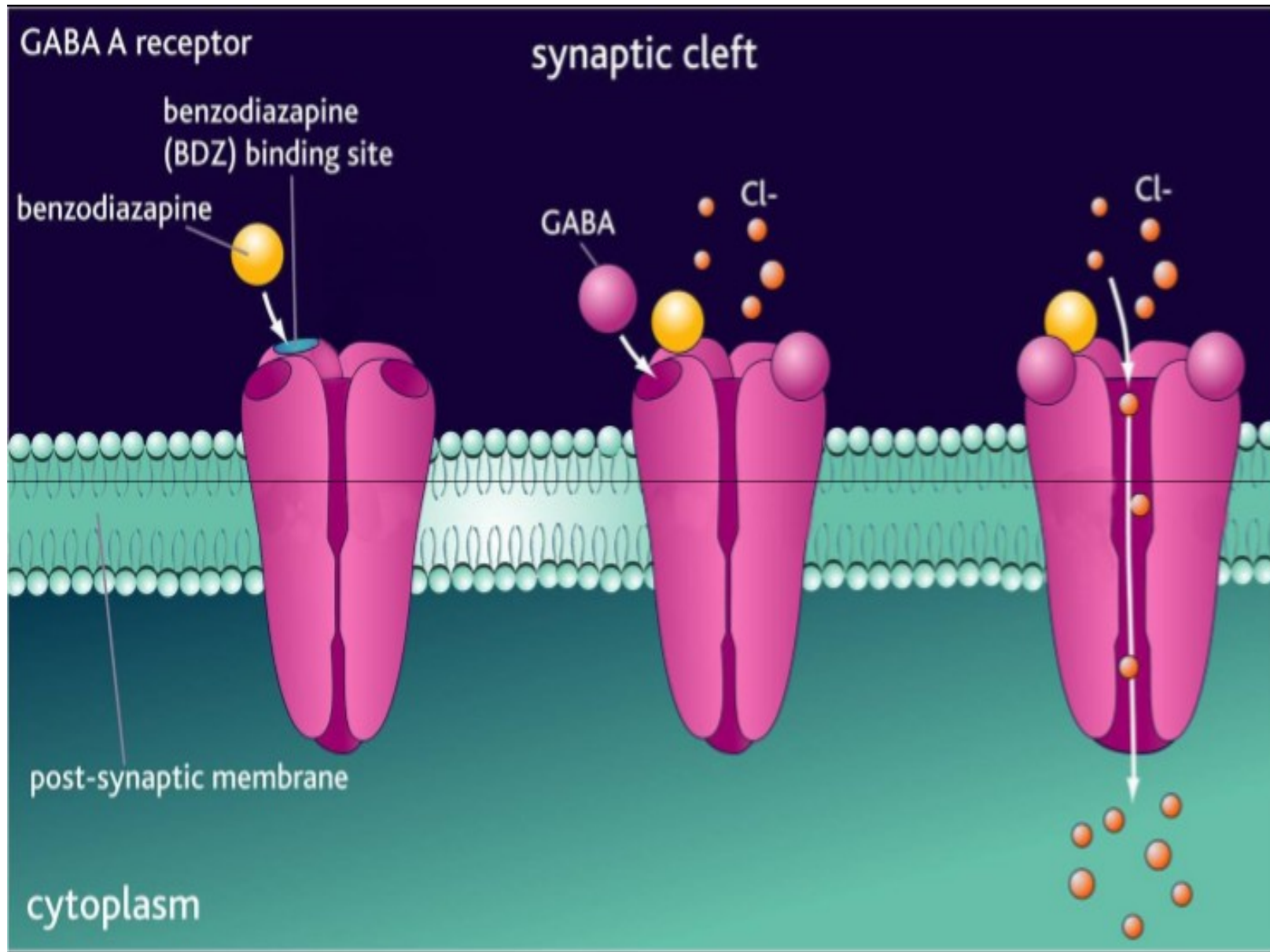
- Benzodiazepine refers to the portion of this structure comprising the benzene ring (A) fused to a seven-member diazepine ring (B).



THE MOLECULAR TARGET FOR BENZODIAZEPINES

- Benzodiazepines act at GABA-A receptors by binding directly to a **specific site that is distinct from the GABA binding site.**
- Benzodiazepines are **allosteric modulators of GABA-A receptor function.**
 - They **increase the affinity of the GABA-A receptor** for GABA and thereby enhance GABA-induced Cl⁻ currents.
 - Benzodiazepines **increase the frequency of opening** of the GABA-A receptor Cl⁻ channel in the presence of GABA.





- **On sleep:** Basic actions
 - Hastens onset of sleep, reduce intermittent awakening and increased total sleep time
- Centrally mediated muscle relaxant –
Clonazepam and Diazepam
- Anticonvulsant: Clonazepam, diazepam and flurazepam



○ Anticonvulsant

- Several benzodiazepines have anticonvulsant activity
- This effect is partially, although not completely, mediated by $\alpha 1$ -GABA-A receptor.
- Parenteral lorazepam is used to suppress the symptoms of delirium tremens.

○ Muscle relaxant

- At high doses, the benzodiazepines relax the spasticity of skeletal muscle
- By increasing presynaptic inhibition in the spinal cord, where the $\alpha 2$ -GABA-A receptors are largely located.



○ Anterograde Amnesia

- Temporary impairment of memory with use of the benzodiazepines
- Mediated by the $\alpha 1$ -GABA-A receptors
- The ability to learn and form new memories is also impaired.
- BDZSs used for initial management of mania and the control of drug-induced hyperexcitability states (eg, phencyclidine intoxication).
- Daytime sedation is more common with BDZSs with slow elimination rates (eg, lorazepam) and those that are



DEPENDENCE

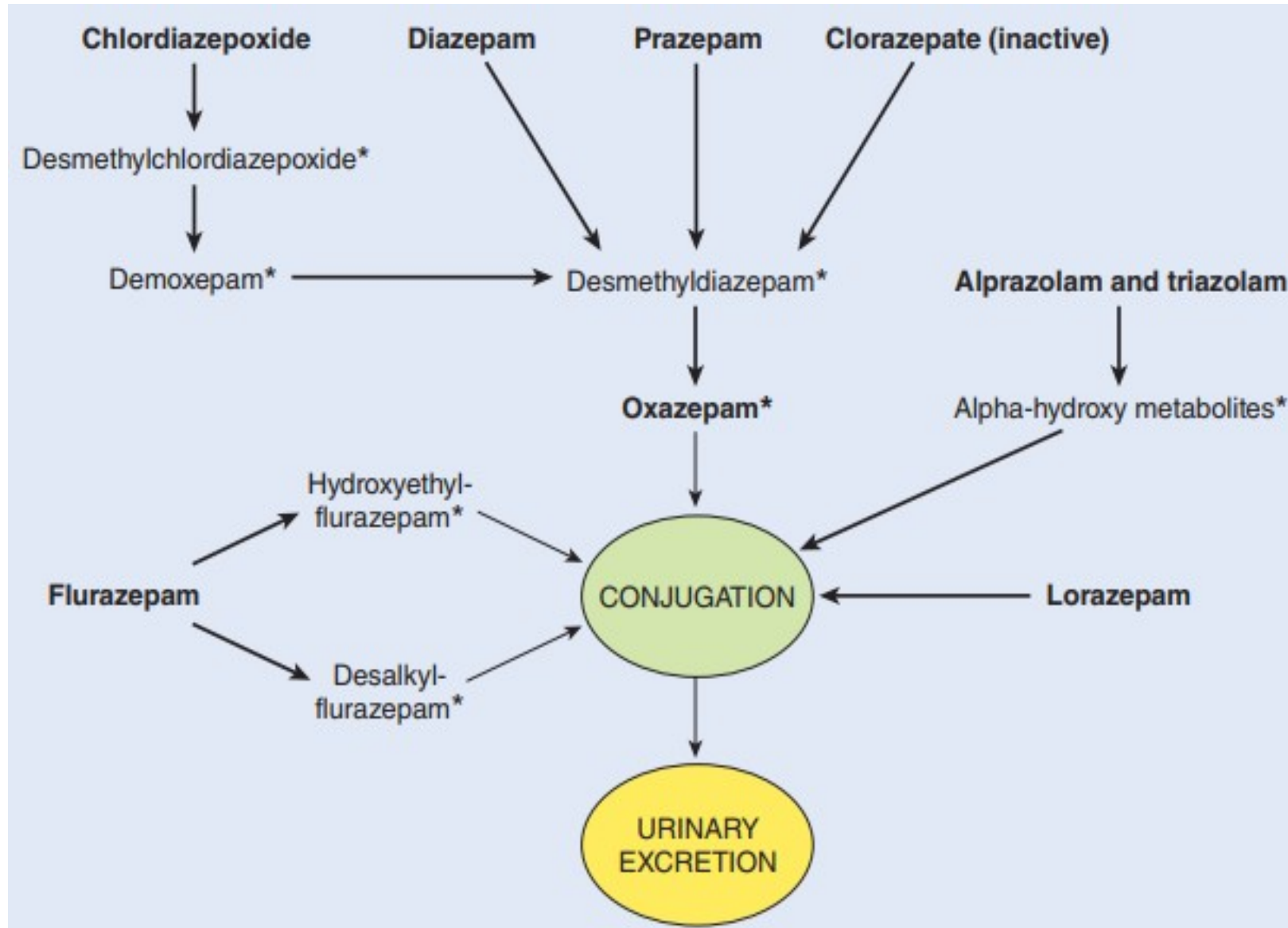
- Psychological and physical dependence can develop if high doses are given for a prolonged period.
- Abrupt discontinuation results in withdrawal symptoms
 - Confusion, anxiety, agitation, restlessness, insomnia, tension, and (rarely) seizures
- Benzodiazepines with a short elimination half life, such as **triazolam**, induce more abrupt and severe withdrawal reactions than the slowly eliminated such as **flurazepam**



PKs

- Marked pharmacokinetic differences – differ in **lipid solubility**
- **Oral absorption** – some rapid, some slow; on IM injection – irregular absorption except lorazepam
- **Plasma protein binding** also varies (Flurazepam – 10%; Diazepam 99%)
- **Metabolism:** In Liver by dealkylation and hydroxylation to metabolites (some are active) – Biological half-life longer than plasma half life
- **Enterohepatic circulation** – Diazepam and excreted in urine as glucoronide conjugate
- BZDs are secreted in milk and crosses placenta





Drugs	Half life (hour)
Diazepam	30 - 60
Nitrazepam	30
Flurazepam	50-100
Alprazolam	12
Chlordiazepoxide	15 - 40
Lorazepam	10 - 20
Temazepam	8 - 12



ADVERSE EFFECT

- Older individuals are more susceptible – Be careful
- Hypnotic doses: Dizziness, vertigo, ataxia, disorientation, amnesia, prolongation of reaction time – impairment of psychomotor skill
- **Hang over** with larger doses – long acting ones
- Weakness, blurring of vision, dry mouth and urinary incontinence
- Tolerance to sedative effect very slowly – little tendency to increase dose – cross tolerance to alcohol and other CNS depressants
- Low withdrawal syndrome – more with ultra rapid ones – anxiety, insomnia, restlessness, malaise, loss of appetite, bad dreams
- **Pregnancy** – flaccidity and respiratory depression in neonate



- **High therapeutic index** – with hypnotic doses – no loss of consciousness or respiratory depression – patient can be aroused
- Hypnotic doses – do not depress CVS or respiratory function.
- No microsomal induction – less drug interaction
- Low abuse liability, mild tolerance, psychological and physical dependence and less withdrawal symptoms



NONBENZODIAZEPINES - ZOPICLONE

- Cyclopyrrolone derivative - active metabolite N-desmethylopiclone
- **MOA**: Binds to α subunit of BZD receptor – hypnotic action
- **Vs BZD**: Sleep resembles but – does not alter REM and tends to prolong stage 3 and stage 4 (Unlike BZD)
- No sleep architecture distortion or withdrawal phenomena
- **Uses**: to wean off insomniacs on BZD and **short term therapy for insomnia**
- **ADRs**: Metallic taste, impairment of judgment and alertness, psychological disturbance – addictive property (rarely)
- Half life: 5 – 6 hours



ZOLPIDEM

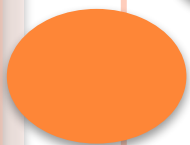
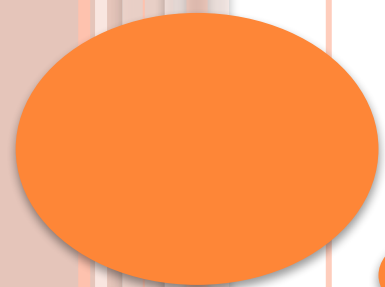
- **MOA:** Acts on $\alpha 1$ subunit of BZD receptor (hypnotic)
- **Actions:** Sleep latency shortened, prolongs sleep time – but **no anticonvulsant, antianxiety or muscle relaxant effects**
 - Lack of effect on sleep stages (REM)
 - Minimal residual day time sedation or fading of effects on repeated use
 - Little rebound insomnia on discontinuation
 - Absence of tolerance, physical dependence and low abuse potential
- **Kinetics:** Completely metabolized in liver – half life –2 hrs
- **Uses:** short term therapy of sleep onset insomnia
- Most popular – even large doses no respiratory depression



ZALEPLON

- **Shortest acting** - acts on $\alpha 1$ subunit of BZD receptor (hypnotic)
- Rapidly absorbed (30% bioavailability – **high first pass**) – rapidly cleared by hepatic metabolism – Half life (1 hr) – no active metabolite
- New class of hypnotics; **rapid onset, short duration**; helps for **falling asleep** but **not for staying asleep**.
- Can be taken late night – no morning sedation, anxiety or insomnia
- No tolerance or dependence
- **Uses:** Sleep-onset insomnia (1-2 weeks therapy)





THE ALCOHOLS

ALCOHOL.

- occupied an important place in the history of humankind for at least 8000 years
- Provides important calories and nutrients ???
- Can relieve anxiety

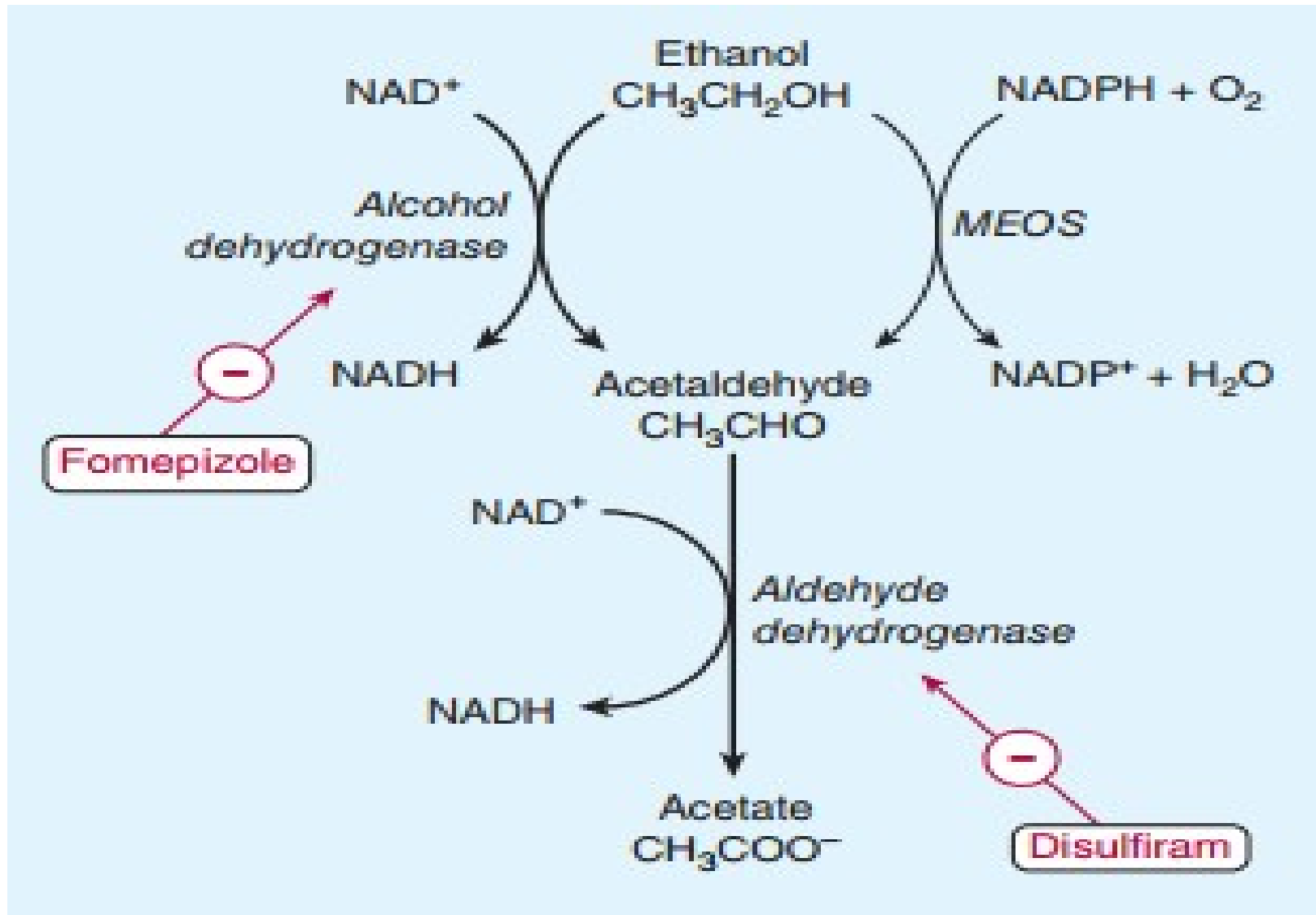


CONT..

- Ethanol is a small water-soluble molecule that is absorbed rapidly from the gastrointestinal tract
- Peak blood alcohol concentrations are reached within 30 minutes.
- Food delays absorption
- $V_d = 0.5-0.7 \text{ L/kg}$



METABOLISM



MANAGEMENT OF ACUTE ALCOHOL INTOXICATION

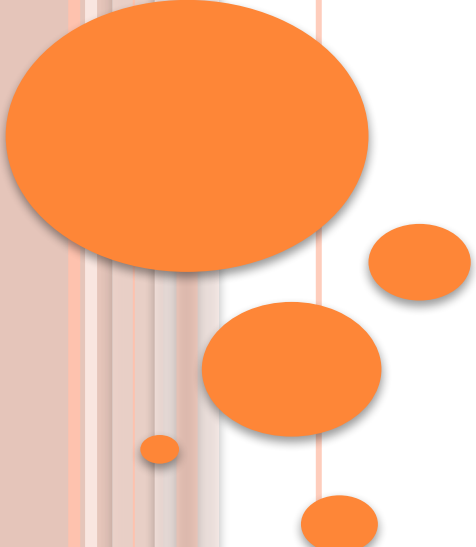
- Prevent severe respiratory depression and aspiration of vomitus
 - Disulfiram,
 - Naltrexone,
 - Acamprosate



Drug-Drug interaction

- Alcohol enhance the hepatotoxic effect of acetaminophen and gastric irritating effects of NSAIDs, thus increasing the risk for development of gastritis and upper GI bleeding.
- Alcohol enhances the sedation effect of antihistamines and produces significant CNS depression.
- Chronic alcoholics interfere with metabolism of certain drugs due to enhanced levels of liver enzyme.





CENTRALLY ACTING MUSCLE RELAXANTS

Skeletal muscle relaxants

Classification

- ❑ **Peripherally acting (Neuromuscular blockers).**
- ❑ **Centrally acting skeletal muscle relaxants**
e.g. Baclofen - Diazepam
- ❑ **Direct acting skeletal muscle relaxants**
e.g. Dantrolene



CENTRALLY ACTING MUSCLE RELAXANTS

- Alleviate muscle spasm& discomfort by acting on CNS
- Ease the stiffness, pain, and spasticity associated with certain muscular conditions.
- They're often used for acute back pain, spinal cord injuries, and conditions like multiple sclerosis.
- Common examples include Baclofen, Cyclobenzaprine, and Tizanidine.



CONT...

- The exact mechanism of action of skeletal muscle relaxants is not well known at this time.
- The use of these drugs may result in a mild general sedative effect producing an overall relaxation.
- Have an onset of action between 30 and 60 minutes
- Duration of action varies among the drugs



CONT..

- 1910-Discovery of phenoxypropanediol
- 1946-Berger and Bradly, incidentally found that mephenesine blocked the nerve conduction in spinal medullary
- 1949-Goodmer's found that the benzimidazole structure also have a myorelaxant activity



CLASSIFICATION

- 1) Propanediol Derivatives
- 2) Benzoxazole Derivatives
- 3) Benzodiazepines
- 4) Baclofen
- 5) Other medicines



DIAZEPAM

- Diazepam acts at GABAA synapses, and its action in reducing spasticity is at least partly mediated in the spinal cord because it is somewhat effective in patients with cord transection.
The initial dosage is 4 mg/d, and it is gradually increased to a maximum of 60 mg/d



BACLOFEN

- p-chlorophenyl-GABA
- Agonist of GABA-B
- Reduces back pain
- plasma half-life of 3–4 hours
- Dosage is started at 15 mg twice daily, increasing as tolerated to 100 mg daily.
- Causes drowsiness and increased seizure activity

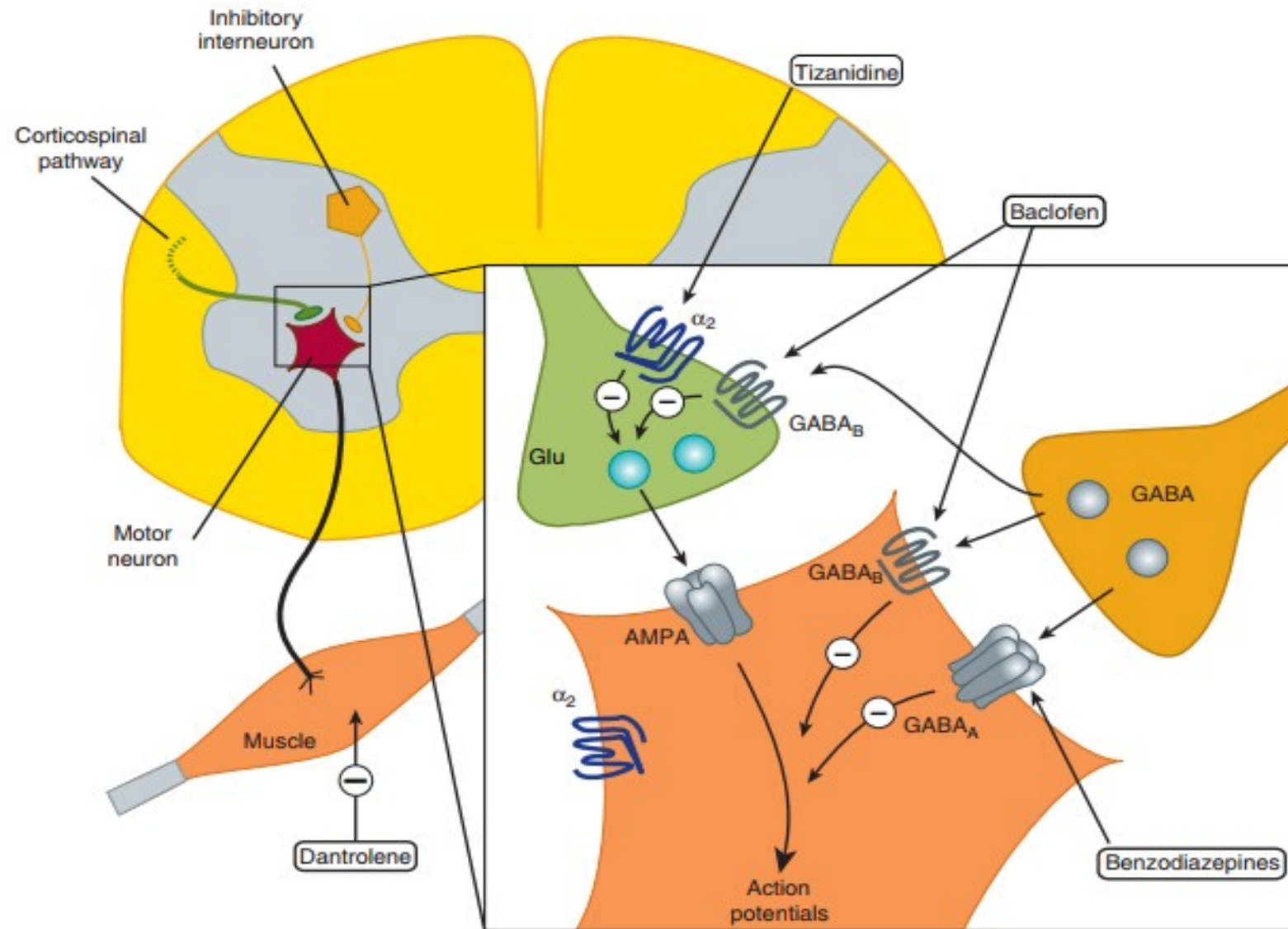


TIZANIDINE



- Congener of clonidine
- Significant agonist effect on α_2 -adrenoceptor
- It reduces muscle spasticity and is assumed to act by increasing presynaptic inhibition of motor neurons.
- Lower effect in BP
- ADR: less muscle weakness, drowsiness, hypotension, dizziness, dry mouth, asthenia, and hepatotoxicity.
- Treatment is initiated at 2 mg every 6–8 hours and can be titrated up to a maximum of 36 mg/d.
- Used for MS and ALS





OTHER CENTRALLY ACTING SPASMOLYTIC DRUGS

- Gabapentin
- Pregabalin



DANTROLENE

- Hydantoin derivative
- Dantrolene interferes with the release of activator calcium through this sarcoplasmic reticulum calcium channel by binding to the RyR1 and blocking the opening of the channel.
- 25 mg daily
- used in the treatment of malignant hyperthermia.

